

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
		(v)		The current through the lamp is smaller than the current through the resistor or values stated 1.2 [A] for the resistor and 1.125 [A] for the lamp (1) [Since the voltage is the same] bigger resistance results in a smaller current (1) Alternative: calculations based $R = 5\ [\Omega]$ for wire (1) and $R = 5.33\ [\Omega]$ for lamp (1) Don't accept reference to gradients		2		2	2	2
		(vi)		Any 2 × (1) from: - At 5.50 V or where the lines cross [the currents are equal so] the powers are equal $[I_{\text{resistor}} \text{ is less than } I_{\text{lamp}}]$ up to 5.50 V, therefore $P_{\text{lamp}} > P_{\text{resistor}}$ (since voltages are the same and $P=VI$) - Beyond 5.50 V, $[I_{\text{resistor}} \text{ is greater than } I_{\text{lamp}} \text{ therefore}] P_{\text{lamp}} < P_{\text{resistor}}$		2		2	2	2
				Question 5 total	3	13	0	16	14	16